

Jinwen (Maggie) Ma

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EDUCATION

University of Toronto

Toronto, ON

Bachelor of Applied Science (B.A.Sc) in Computer Engineering + PEY Co-op Sept. 2024 – May 2029 (Expected)

- **Relevant courses:** Digital Systems (Verilog), Operating Systems, Computer Hardware, Digital Electronics, Introduction to Control Systems.

EXPERIENCE

Development Engineer Intern

May 2026 – Aug. 2026

Tianjin Shenzhou General Data Technology Co., Ltd.

Beijing, China

- Wrote SQL queries to retrieve, validate, update, and organize test data for application development and quality assurance.
- Tested client-facing applications, documented reproducible bugs, and retested corrected features to verify fixes.
- Created user and testing manuals documenting application workflows, test procedures, expected outcomes, and troubleshooting steps.

Electronics Sub Team – Team Member

Sep. 2025 – Present

U of T Human Powered Vehicles Design Team (HPVDT)

Toronto, ON

- Designed and prototyped control circuits using breadboards and KiCad to support PCB schematic and layout development for vehicle electronics.
- Integrated and debugged sensors, microcontrollers, and PCB hardware over I²C and SPI while collaborating with mechanical and software teams.

PROJECTS

AutoFlora Smart Plant Care System

Aug. 2026 – Present

Embedded Systems Personal Project – Developer

Toronto, ON

- Developing an STM32-based plant-care system in C/C++ to monitor soil moisture, temperature, humidity, and ambient light using ADC and I²C sensors.
- Integrated AHT20 and BH1750 sensors over I²C with UART logging, debugging device addressing, initialization, wiring, and measurement conversion.
- Implementing calibrated soil-moisture sensing and a MOSFET-driven watering system with float-switch lockout, pulse watering, cooldown control, and fault protection.

CitySense GIS Map Development

Jan. 2026 – May 2026

Engineering Project, U of T – Developer

Toronto, ON

- Developed an interactive C++ GIS displaying streets, intersections, buildings, POIs, and navigation routes across large map datasets.
- Implemented A* pathfinding with estimated travel times, highlighted driving/walking routes, and turn-by-turn directions.
- Built zoom-based level-of-detail rendering, label collision detection, real-time street search, POI filtering, intersection lookup, and customizable themes.

Brick Breaker Game Development on FPGA (DE1-SoC Board)

Oct. 2025 – Nov. 2025

Digital Systems Project, U of T – Developer

Toronto, ON

- Implemented a Brick Breaker game in Verilog with real-time VGA output on the DE1-SoC FPGA.
- Developed ball movement, collision detection, paddle control, and score logic synchronized with the 50 MHz system clock.
- Debugged timing and display behaviour using ModelSim, applying finite-state-machine control and modular Verilog design.

TECHNICAL SKILLS

Programming: C, C++, Python, JavaScript, TypeScript, HTML/CSS, Verilog, RISC-V Assembly

Software & Tools: Git, MATLAB, Quartus Prime, ModelSim, DESim, KiCad, STM32CubeMX, CMake, VS Code

Hardware & Systems: STM32, FPGA Design, Embedded Systems, I²C, SPI, UART, ADC, Sensor Interfacing, Circuit Analysis, PCB Design